CLASS 8 — MATHS · EASY LEVEL CHAPTER TEST SET 2

Chapter 10: **Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a pie chart a sector represents 15% of the data. Its central angle (in degrees) is: 1

(a) 54 **(b)** 45 **(c)** 60 **(d)** 30

2. 6 workers can finish a job in 10 days. The number of days 12 workers (working at the same rate) need is: 1

(a) 60 **(b)** 22 **(c)** 7 **(d)** 5

3. The ratio 24:36 in its simplest form is: 1

(a) 2:3 **(b)** 2:4 **(c)** 3:2 **(d)** 3:3

4. If 4:7 = 12:x, then x is: 1

(a) 24 **(b)** 21 **(c)** 10 **(d)** 19

5. ₹ 550 is divided between two friends in the ratio 2:3. The larger share (in ₹) is: 1

(a) 350 **(b)** 220 **(c)** 330 **(d)** 275

6. On a map, 1 cm represents 25 km. A distance of 4 cm on the map is actually (in km): 1

(a) 100 **(b)** 29 **(c)** 40 **(d)** 125

7. In a pie chart a sector represents 20% of the data. Its central angle (in degrees) is: 1

(a) 60 **(b)** 40 **(c)** 72 **(d)** 80

8. 8 workers can finish a job in 15 days. The number of days 20 workers (working at the same rate) need is: 1

(a) 35 **(b)** 120 **(c)** 6 **(d)** 8

9. The ratio 35:49 in its simplest form is: 1

(a) 6:7 **(b)** 5:8 **(c)** 7:5 **(d)** 5:7

10. If 5:6 = 25:x, then x is: 1

(a) 11 **(b)** 31 **(c)** 30 **(d)** 35

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. 15 men can build a wall in 16 days. In how many days can 12 men build it? 2

12. Divide ₹ 650 between A and B in the ratio 6:7. 2

13. If 5 pens cost ₹ 175, find the cost of 12 pens. 2

14. 6 men can build a wall in 20 days. In how many days can 15 men build it? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. The scale of a map is 1 cm = 8 km. Two cities are 9 cm apart on the map. Find the actual distance. Also find the map distance of two towns that are actually 56 km apart. 3

16. Divide ₹ 1500 among A, B and C in the ratio 3:5:7. 3

17. In a pie chart of 150 students' favourite games, cricket takes up 40% of the chart. Find the central angle of the cricket sector and the number of students who like cricket. 3

OR

The scale of a map is 1 cm = 40 km. Two cities are 6 cm apart on the map. Find the actual distance. Also find the map distance of two towns that are actually 280 km apart. 3

18. Divide ₹ 1800 among A, B and C in the ratio 2:3:4. 3

OR

In a pie chart of 400 students' favourite games, cricket takes up 15% of the chart. Find the central angle of the cricket sector and the number of students who like cricket. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. The data of 50 students' choices is: Bus 20, Car 14, Cycle 8, Walk 8. Find the central angle of each sector for a pie chart and verify that the angles add to 360°. 5

OR

A camp has food for 24 people for 10 days. If 6 more people join the camp, for how many days will the same food last? 5

20. The data of 30 students' choices is: Red 10, Blue 8, Green 6, Yellow 6. Find the central angle of each sector for a pie chart and verify that the angles add to 360°. 5

OR

A camp has food for 45 people for 8 days. If 15 more people join the camp, for how many days will the same food last? 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND